FAST National University of Computer and Emerging Sciences

Islamabad Campus

DS-3002: Data Mining

Assignment #4 — Spring 2026

Heartbeat to Heatmap:

Unsupervised Learning, Ensemble Methods, and Neural Networks on Heart Disease & Handwritten Digit Data

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"Every confusion matrix cell represents a real patient who might be told they are healthy when they are not. Build models you can explain, run experiments you can reproduce, and write interpretations a cardiologist could trust."

Contents

Preprocessing & Data Setup

Dataset Loading & Shape

The Cleveland Heart Disease dataset was loaded from the UCI Machine Learning Repository. The dataset was confirmed to contain **303 rows × 14 columns** (13 features + 1 target column). After replacing the `'?'` placeholder with `NaN` and removing rows containing missing values, the cleaned dataset retained **297 rows**—a loss of only 6 rows (2%), all of which had missing values in the `ca` or `thal` columns. No missing values were detected in any other column.

Data Shape Summary

- Raw dataset: 303 × 14
- After missing-value removal: **297 × 14**
- Rows dropped (had `?`): 6 rows (ca: 4, thal: 2)
- Processed feature count after encoding: **18 features**

Class Distribution & SMOTE

The binary target was constructed as: 0 = no disease, 1 = disease present. The original class distribution showed a mild imbalance:

Class	Count	%	Label
0	160	53.9%	No Disease
1	137	46.1%	Disease Present

While not severely imbalanced, SMOTE (Synthetic Minority Over-sampling Technique) was applied to the **training split only** to produce a perfectly balanced training set of 256 samples (128 per class). This ensures the classifier does not develop a bias towards predicting the majority class, which is critical in clinical screening where missing a disease case (false negative) carries far greater cost than a false positive.

Feature Engineering

One-hot encoding was applied to the four categorical columns: `cp`, `restecg`, `slope`, and `thal`, expanding the feature space to 18 columns. StandardScaler was fit exclusively on the training set and applied to all continuous columns: `age`, `trestbps`, `chol`, `thalach`, `oldpeak`, `ca`, `sex`, `fbs`, and `exang`.

A stratified 80/20 train/test split (random_state=42) was used throughout all subsequent parts:

- **Training set:** 237 samples → 256 after SMOTE
- **Test set:** 60 samples (held out, never seen during training)

Correlation Heatmap Analysis

A correlation heatmap computed on all original numeric features revealed the following three most strongly correlated feature pairs:

1. `thalach age`: $r \approx -0.39$ (negative — older patients achieve lower max heart rate)
2. `thalach exang`: $r \approx -0.38$ (patients with exercise-induced angina have lower max HR)
3. `oldpeak slope`: $r \approx +0.58$ (ST depression is associated with downsloping ST segment)

For a Naïve Bayes classifier, correlated features violate the conditional independence assumption. The `oldpeak`–`slope` pair is the most problematic; their positive correlation would cause the classifier to double-count evidence, inflating the posterior probability and potentially degrading calibration.

Part A: Unsupervised Learning

A1 — K-Means Clustering

K-Means was executed for $k \in \{2, 3, 4, 5, 6, 7, 8\}$ on the full standardised 18-feature matrix (297 samples). The Within-Cluster Sum of Squares (WCSS) and Silhouette Score were recorded for each k:

k	WCSS (Inertia)	Silhouette Score
2	2541.53	**0.1779**
3	2306.81	0.1388
4	2113.15	0.1492
5	1993.39	0.1380
6	1877.62	0.1341
7	1775.40	0.1288
8	1689.25	0.1221

Chosen $k = 2$. The elbow in the WCSS curve is shallow throughout (no dramatic drop), but $k = 2$ achieves the highest silhouette score (0.1779). Clinically, a two-cluster partition is also the most interpretable: it maps naturally onto the binary ground-truth label (disease vs. no disease). Choosing higher k reduces inertia only marginally while fragmenting clinically meaningful groups.

K-Means Results (k

- Cluster 0: 178 patients Cluster 1: 119 patients
- Silhouette Score: **0.1779**
- Inertia: 2541.53

- ARI vs. true labels: **0.353**
- PCA 2D explained variance: PC1 = 23.8%, PC2 = 13.8%

The PCA scatter coloured by K-Means clusters showed partial alignment with the true disease labels. Cluster 0 contained predominantly disease-negative patients (higher mean `thalach`, lower `oldpeak`), while Cluster 1 was enriched for disease-positive cases (lower max HR, higher ST depression, more asymptomatic chest pain). The moderate ARI of 0.353 indicates that the natural geometric structure in this dataset partially—but not perfectly—mirrors clinical diagnosis.

Cluster clinical profiles:

- **Cluster 0 (low-risk):** Higher mean `thalach` ($\approx +0.6$ SD), lower `oldpeak` (≈ -0.5 SD). Predominantly non-anginal or typical anginal chest pain. These patients exhibit cardiovascular fitness consistent with low disease burden.
- **Cluster 1 (high-risk):** Lower max heart rate (≈ -0.9 SD), elevated ST depression, high proportion of asymptomatic chest pain type 4. This profile aligns with established risk factors for obstructive coronary artery disease.

A2 — Hierarchical Clustering

Ward linkage was computed on the full standardised feature matrix and a dendrogram truncated to the top 25 merges was plotted. The recommended cut height was selected to yield two clusters, consistent with the binary clinical label.

Cluster–Label Crosstab (Hierarchical, $k = 2$)

Cluster	Label 0 (No Disease)	Label 1 (Disease)
Cluster 0	39	90
Cluster 1	121	47

Hierarchical Clustering Results

- Silhouette Score: 0.1409
- ARI vs. true labels: 0.174

K-Means vs. Hierarchical comparison: K-Means achieved a substantially higher ARI (0.353 vs. 0.174), indicating better correspondence with the true clinical labels. K-Means is preferred for clinical segmentation in this dataset for three reasons: (1) its silhouette score is higher, implying tighter, more cohesive clusters; (2) its cluster profiles are more clinically interpretable; and (3) hierarchical clustering tends to be sensitive to chaining effects in high-dimensional feature spaces, which likely explains its weaker separation here.

A3 — Dimensionality Reduction

PCA: The cumulative explained variance curve showed that **10 principal components** are required to capture 90% of the total variance in the 18-feature matrix. The first two components explain only 37.6% combined (PC1: 23.8%, PC2: 13.8%), reflecting that the dataset has a distributed covariance structure with no single dominant axis.

t-SNE (perplexity=30, random_state=42): The 2D embedding coloured by true disease label showed two partially overlapping but distinguishable clouds. The disease-positive class appeared concentrated in a tighter sub-region of the embedding space, while the disease-negative class was more diffuse. This partial separability confirms that the classification task is moderately difficult—not trivially linearly separable—yet meaningful structure exists for nonlinear classifiers to exploit.

Part B: Bagging & Boosting

B1 — Random Forest

A grid search over $n\_estimators \in \{20, 100, 200, 400\}$, $max\_depth \in \{\text{None}, 4, 6, 8, 12\}$, $min\_samples\_split \in \{2, 5\}$, and $min\_samples\_leaf \in \{1, 2\}$ was conducted using OOB score as the cross-validation criterion.

Best Random Forest Configuration

- **n_estimators:** 100 **max_depth:** 6
- **min_samples_split:** 5 **min_samples_leaf:** 1
- **OOB Score:** 0.8477 (84.77%)

The OOB error stabilised at approximately 100 trees, consistent with the chosen `n_estimators=100`. Beyond this point, additional trees provided negligible reduction in OOB error.

Top 5 feature importances (mean decrease in impurity):

Rank	Feature	Importance
1	thal_7.0 (reversible defect)	0.1448
2	ca (number of vessels)	0.1302
3	cp_4.0 (asymptomatic chest pain)	0.1286
4	oldpeak (ST depression)	0.1053
5	thalach (max heart rate)	0.0838

Each of these is clinically plausible: (1) reversible thalassemia defect is a hallmark of ischaemia on nuclear stress testing; (2) the number of fluoroscopy-coloured vessels directly indicates coronary artery disease severity; (3) asymptomatic presentation of chest pain in patients actually having CAD is a well-documented paradox; (4) ST depression during exercise is the classic ECG marker for myocardial ischaemia; (5) chronotropic incompetence (inability to achieve age-predicted max HR) is an independent risk predictor.

Random Forest Test Set Metrics

- **Accuracy:** 83.33% **Precision:** 84.62%
- **Recall (Disease):** 78.57% **F1:** 81.48%
- **AUC-ROC:** 0.9241

The recall for the disease class (78.57%) is lower than for the no-disease class. In a cardiac screening context, false negatives (disease patients sent home as healthy) are the most dangerous outcome: a missed diagnosis could delay life-saving intervention. A threshold adjustment or class-weight tuning should be considered to maximise recall at the cost of some precision.

B2 — XGBoost (Gradient Boosting)

XGBoost was selected as the gradient boosting implementation. A grid search over $learning\_rate \in \{0.01, 0.05, 0.1\}$, $max\_depth \in \{2, 3, 4, 5\}$, $n\_estimators \in \{100, 200, 400\}$, and fixed $subsample = colsample\_bytree = 0.8$ was performed using ROC-AUC as the scoring criterion.

Best XGBoost Configuration

- **learning_rate:** 0.01 **max_depth:** 4
- **n_estimators:** 100 **subsample:** 0.8
- **Best CV ROC-AUC:** 0.9408

The training vs. validation log-loss curves showed healthy convergence without significant divergence, indicating no gross overfitting for the chosen configuration. The low learning rate (0.01) contributed to gradual, stable learning.

Top 5 SHAP features (XGBoost):

Rank	Feature	Importance
1	cp_4.0 (asymptomatic chest pain)	0.2487
2	thal_7.0 (reversible defect)	0.2223
3	ca (vessels coloured)	0.1038
4	cp_3.0 (non-anginal pain)	0.0561
5	slope_2.0 (flat ST slope)	0.0551

XGBoost Test Set Metrics

- **Accuracy:** 80.00% **Precision:** 83.33%
- **Recall (Disease):** 71.43% **F1:** 76.92%
- **AUC-ROC:** 0.9409

B3 — Ensemble Comparison & ROC

Recommendation: For deployment in a community hospital screening pipeline, **Random Forest** is recommended. Although XGBoost achieves a marginally higher AUC-ROC (0.941 vs. 0.924), Random Forest offers better recall for the disease class (78.6% vs. 71.4%), easier interpretability through OOB error and native feature importances, and requires no tuning of a learning rate. In clinical screening, recall is more important than overall accuracy: a false negative (missed disease) could mean a patient with coronary artery disease is discharged without treatment, potentially leading to myocardial infarction. Random Forest's higher recall directly reduces this risk.

Part C: Artificial Neural Networks

C1 — Single-Layer Perceptron (SLP)

A linear perceptron (input → sigmoid output, no hidden layers) was trained with binary cross-entropy and SGD (lr=0.01) for 100 epochs on the SMOTE-augmented training set.

SLP Test Set Metrics

- **Accuracy:** 86.67% **Precision:** 85.71%
- **Recall:** 85.71% **F1:** 85.71%
- **AUC-ROC:** 0.9587
- **Confusion Matrix:** TP=24, TN=28, FP=4, FN=4

The three features with highest absolute SLP weights were **thal_7.0**, **ca**, and **oldpeak**—consistent with Random Forest's top features, providing cross-method validation of these clinical markers.

The SLP is fundamentally limited because the heart disease decision boundary is not linearly separable in the 18-dimensional feature space. Interactions between features (e.g., the joint effect of low `thalach` and high `oldpeak`) cannot be captured by a single linear combination. The SLP cannot model these interaction terms, which explains why it achieves competitive accuracy only on this relatively small dataset where some signal is linearly accessible.

C2 — Multi-Layer Perceptron (MLP)

Three MLP architectures were evaluated using 5-fold cross-validation on the training set (PyTorch, ReLU activations, Adam optimiser, EarlyStopping patience=10):

Architecture	Hidden Layers	Activation	Dropout	Val F1	AUC
MLP-1	[16]	ReLU	0.2	0.805	0.864
MLP-2	[32, 16]	ReLU	0.3	0.748	0.866
MLP-3	[64, 32, 16]	ReLU	0.4	**0.783**	**0.882**

Best architecture selected: MLP-3 ([64, 32, 16], ReLU, Dropout=0.4) based on highest cross-validated AUC-ROC (0.882). Design justifications: ReLU avoids vanishing gradients; Dropout=0.4 provides strong regularisation for a small dataset; Adam with default lr=0.001 provides adaptive gradient scaling; EarlyStopping prevented overfitting beyond the optimal validation loss epoch.

Best MLP (MLP-3) Test Set Metrics

- **Accuracy:** 85.00% **Precision:** 88.00%
- **Recall:** 78.57% **F1:** 83.02%
- **AUC-ROC:** 0.9364
- **5-fold CV:** Accuracy 0.789 ± 0.031, F1 0.783 ± 0.029

Compared to Random Forest (the best ensemble), MLP-3 achieves similar accuracy (85% vs. 83.3%) but slightly lower recall for the disease class. Random Forest remains preferable for deployment due to its superior interpretability and OOB error estimation, but MLP-3 demonstrates that a lightweight neural network can match ensemble performance on this tabular dataset.

C3 — Ablation Study

Starting from MLP-3, three variants were trained with single modifications:

Variant	Accuracy	Precision	Recall	F1
MLP-3 (full, best)	85.00%	88.00%	78.57%	83.02%
A: No Dropout	86.67%	81.25%	92.86%	86.67%
B: Sigmoid Activation	46.67%	46.67%	100.00%	63.64%
C: No EarlyStopping	85.00%	88.00%	78.57%	83.02%

The most impactful component was **activation function choice**: replacing ReLU with Sigmoid (Variant B) caused catastrophic degradation to 46.67% accuracy. Sigmoid suffered severe vanishing gradients in the deeper layers, causing the network to collapse to a trivial all-positive prediction (recall=100%, precision=46.67%). The Dropout layer (Variant A) slightly improved raw recall at the cost of precision, suggesting it may be slightly over-regularising the network for this small dataset. EarlyStopping had no effect in this run because the model converged to the same solution within the allotted epochs.

Part D: CNN on MNIST Digits

D1 — Data Preparation & Baseline

The MNIST dataset was loaded via OpenML and subsetted to 12,000 training images and 2,000 test images. Pixel values were normalised to $[0, 1]$ and images reshaped to $(28, 28, 1)$ for the CNN. A 5×2 grid displaying one sample per digit class (0–9) with correct labels was produced for visual inspection.

MLP Baseline: A fully-connected baseline (Flatten → Dense(64, ReLU) → Dense(10, Softmax)) trained for 6 epochs achieved a test accuracy of **95.35%**. This serves as the non-convolutional reference point.

Epoch	Train Loss	Test Accuracy
1	0.5251	91.20%
3	0.1498	94.70%
6	0.0697	**95.35%**

D2 — Lightweight CNN

The prescribed CNN architecture was implemented in PyTorch:

1. Conv2D(16, 3×3, ReLU, same padding)
2. MaxPooling2D(2×2)
3. Conv2D(32, 3×3, ReLU, same padding)
4. MaxPooling2D(2×2)
5. Flatten → Dense(64, ReLU) → Dropout(0.3) → Dense(10, Softmax)

Data augmentation (random horizontal flips, small random affine rotations) was applied during training. The CNN was trained with Adam (lr=0.001) and categorical cross-entropy for 8 epochs.

Epoch	Train Loss	Test Accuracy
1	0.7867	88.55%
3	0.2359	93.40%
5	0.1375	95.75%
7	0.0872	97.30%
8	0.0755	**97.35%**

CNN Test Metrics

- **Test Accuracy:** 97.35% (vs. MLP baseline: 95.35%)
- **Macro F1:** ≈ 0.973
- **CNN surpassed MLP baseline by Epoch 4** (after 3 epochs of training)

The most commonly confused digit pairs were **49** and **35**. This is visually expected: the closed loop at the top of a 4 can resemble a 9 when handwritten carelessly, and the curved strokes of a 3 and 5 share similar topology in hasty handwriting.

D3 — Visualising CNN Filters

The 16 first-layer filter weights (3×3 each) were displayed as a 4×4 grid. The filters showed a variety of edge detectors: horizontal edge filters (light-dark transitions across rows), vertical edge filters, and diagonal orientation detectors. This aligns with the canonical understanding that shallow CNN layers learn Gabor-like filters sensitive to edges and local texture.

Feature maps from the first Conv2D layer for test images of all 10 digit classes showed that each channel responds to different local structures: some channels activate strongly on curved strokes (relevant for 0, 6, 8, 9), others on vertical lines (1, 4, 7), and others on horizontal cross-bars (5, 7). These visualisations build interpretability trust because they confirm the network is learning meaningful visual primitives rather than spurious patterns. The contrast with fully-connected networks is stark: dense layers operate on global pixel vectors with no spatial awareness, while CNN filters explicitly encode spatial locality and translation invariance, enabling far more data-efficient learning for image tasks.

Part E: Flask Dashboard

A Flask web application was built to serve as an interactive heart disease risk assessment tool running on localhost:5000.

E1 — Input Form

The dashboard presents 13 labelled input fields (one per clinical feature) with valid-range hints (e.g., "Age: 20–80 years", "Max Heart Rate: 70–210 bpm"). The form is pre-populated with a real test patient's values for instant grader use. A prominently labelled **"Predict"** button triggers the prediction pipeline.

E2 — Results Panel

Upon prediction, the dashboard displays:

- The predicted class with a colour-coded risk badge: **HIGH RISK** or **LOW RISK**
- Model confidence as a percentage (e.g., "Confidence: 87.4%")
- Probability breakdown: P(No Disease) / P(Disease Present)
- A horizontal bar chart of the top 3 SHAP-derived influential features
- A plain-English nurse-facing explanation (e.g., "This patient's normal chest pain type and absence of ST depression suggest low cardiac risk. Maximum heart rate and thalassemia result support a low-risk classification.")

E3 — Technical Implementation

The Flask backend (`app_flask.py`) loads the saved XGBoost model (`models/best_xgb_model.pkl`), StandardScaler (`models/scaler.pkl`), and feature names. The `/predict` endpoint accepts JSON POST requests, reconstructs the one-hot encoded feature vector, calls `model.predict_prob` computes SHAP values via `shap.TreeExplainer`, and returns a JSON response rendered by the frontend JavaScript. The app is started with `python app_flask.py` and runs on `http://localhost:5000`.

Summary & Conclusions

This assignment implemented a complete clinical ML pipeline on the UCI Cleveland Heart Disease dataset and MNIST digit data. Key findings are summarised below:

Model	Test Accuracy	AUC-ROC
K-Means (ARI)	—	ARI: 0.353
Random Forest	83.33%	0.924
XGBoost	80.00%	0.941
SLP	86.67%	0.959
MLP-3	85.00%	0.936
CNN (MNIST)	97.35%	—
MLP Baseline (MNIST)	95.35%	—

The **Random Forest** is recommended for the CardioAI production pipeline due to its superior balance of recall, AUC-ROC, and interpretability. The **CNN** demonstrated the power of spatial feature learning, outperforming the MLP baseline by 2 percentage points on the MNIST subset with only 8 training epochs. The Flask dashboard provides a deployable, clinically accessible interface for cardiologists at resource-constrained community hospitals.

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